

## labwork 8

October 5, 2026

## 0.1 RGB to HSV

The goal of the exercise is to turn an image from RGB format to HSV and then to turn it back to RGB.

### 0.1.1 Preparation

First, we normalize the RGB values from  $[0, 255]$  to  $[0, 1]$ . Then, we take the max, the min and the delta between min and max.

```
r = src[y, x, 0] / 255
g = src[y, x, 1] / 255
b = src[y, x, 2] / 255
maxV = max(r, g, b)
minV = min(r, g, b)
delta = maxV - minV
```

### 0.1.2 Conversion

To turn our RGB image to HSV we apply the following formula:

$$H = \begin{cases} 0^\circ & \Delta = 0 \\ 60^\circ \times (\frac{G-B}{\Delta} \bmod 6) & \max = R \\ 60^\circ \times (\frac{B-R}{\Delta} + 2) & \max = G \\ 60^\circ \times (\frac{R-G}{\Delta} + 4) & \max = B \end{cases}$$
$$S = \begin{cases} 0 & \max = 0 \\ \frac{\Delta}{\max} & \max \neq 0 \end{cases}$$
$$V = \max$$

Which gives in python:

```
if (maxV == 0):
    dst[1, y, x] = maxV
else:
    dst[1, y, x] = delta / maxV

dst[2, y, x] = maxV
if delta == 0:
    dst[0, y, x] = 0
elif maxV == r:
    dst[0, y, x] = 60 * (((g - b) / delta) % 6)
elif maxV == g:
    dst[0, y, x] = 60 * ((b - r) / delta + 2)
else:
    dst[0, y, x] = 60 * ((r - g) / delta + 4)
```

## 0.2 HST to RGB

### 0.2.1 Preparation

To turn the HSV image back to RGB we simply implement the given formulas:

```
d = h / 60
f = d - int(d)
l = v * (1 - s)
m = v * (1 - f * s)
n = v * (1 - (1 - f) * s)
```

### 0.2.2 Conversion

Then we compute the right result depending on the value of H going from 0 to 360.

```
if (0 <= h < 60):
    dst[y, x, 0] = round(v * 255)
    dst[y, x, 1] = round(n * 255)
    dst[y, x, 2] = round(l * 255)
elif (60 <= h < 120):
    dst[y, x, 0] = round(m * 255)
    dst[y, x, 1] = round(v * 255)
    dst[y, x, 2] = round(l * 255)
```